



5.3.2 Permutation matrices ¶

5.3.2 Permutation matrices, part 1

fit width



Recall that we already discussed permutation in [Subsection 4.4.4](#) in the setting of column pivoting when computing the QR factorization.

Definition 5.3.2.1. Given

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix},$$

where $\{\pi_0, \pi_1, \dots, \pi_{n-1}\}$ is a permutation (rearrangement) of the integers $\{0, 1, \dots, n-1\}$, we define the permutation matrix $P(p)$ by

$$P(p) = \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}.$$

Homework 5.3.2.1. Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix}.$$

Evaluate $P(p)x$.

▼ [Solution](#)

$$\begin{aligned} P(p)x &= \text{< definition >} \\ &= \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix} x \\ &= \text{< matrix-vector multiplication by rows >} \\ &= \begin{pmatrix} e_{\pi_0}^T x \\ \vdots \\ e_{\pi_{n-1}}^T x \end{pmatrix} \\ &= \text{< } e_j^T x = x_j \text{ >} \\ &= \begin{pmatrix} \chi_{\pi_0} \\ \vdots \\ \chi_{\pi_{n-1}} \end{pmatrix} \end{aligned}$$

The last homework shows that applying $P(p)$ to a vector x rearranges the elements of that vector according to the permutation indicated by the vector p .

Homework 5.3.2.2. Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{n-1}^T \end{pmatrix}.$$

Evaluate $P(p)A$.

▼ [Solution](#)

$$\begin{aligned}
P(p)A &= \text{< definition >} \\
&= \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix} A \\
&= \text{< matrix-matrix multiplication by rows >} \\
&= \begin{pmatrix} e_{\pi_0}^T A \\ \vdots \\ e_{\pi_{n-1}}^T A \end{pmatrix} \\
&= \text{< } e_j^T A = \tilde{a}_j^T \text{ >} \\
&= \begin{pmatrix} \tilde{a}_{\pi_0}^T \\ \vdots \\ \tilde{a}_{\pi_{n-1}}^T \end{pmatrix}
\end{aligned}$$

The last homework shows that applying $P(p)$ to a matrix A rearranges the rows of that matrix according to the permutation indicated by the vector p .

Homework 5.3.2.3. Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = (a_0 \quad \cdots \quad a_{n-1}).$$

Evaluate $AP(p)^T$.

▼ [Solution](#)

$$\begin{aligned}
AP(p)^T &= \text{< definition >} \\
&= A \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}^T \\
&= \text{< transpose } P(p) \text{ >} \\
&= A (e_{\pi_0} \quad \cdots \quad e_{\pi_{n-1}}) \\
&= \text{< matrix-matrix multiplication by columns >} \\
&= (Ae_{\pi_0} \quad \cdots \quad Ae_{\pi_{n-1}}) \\
&= \text{< } Ae_j = a_j \text{ >} \\
&= (a_{\pi_0} \quad \cdots \quad a_{\pi_{n-1}})
\end{aligned}$$

The last homework shows that applying $P(p)^T$ from the right to a matrix A rearranges the columns of that matrix according to the permutation indicated by the vector p .

Homework 5.3.2.4. Evaluate $P(p)P(p)^T$.

► [Answer](#)

▼ [Solution](#)

$$\begin{aligned}
P(p)P(p)^T &= \text{< definition >} \\
&= \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix} \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}^T \\
&= \text{< transpose } P(p) \text{ >} \\
&= \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix} (e_{\pi_0} \quad \cdots \quad e_{\pi_{n-1}}) \\
&= \text{< evaluate >} \\
&= \begin{pmatrix} e_{\pi_0}^T e_{\pi_0} & e_{\pi_0}^T e_{\pi_1} & \cdots & e_{\pi_0}^T e_{\pi_{n-1}} \\ e_{\pi_1}^T e_{\pi_0} & e_{\pi_1}^T e_{\pi_1} & \cdots & e_{\pi_1}^T e_{\pi_{n-1}} \\ \vdots & \vdots & \ddots & \vdots \\ e_{\pi_{n-1}}^T e_{\pi_0} & e_{\pi_{n-1}}^T e_{\pi_1} & \cdots & e_{\pi_{n-1}}^T e_{\pi_{n-1}} \end{pmatrix} \\
&= \text{< } e_i^T e_j = \cdots \text{ >} \\
&= \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}
\end{aligned}$$

5.3.2 Permutation matrices, part 2

fit width



We will see that when discussing the LU factorization with partial pivoting, a permutation matrix that swaps the first element of a vector with the π -th element of that vector is a fundamental tool.

Definition 5.3.2.2. Elementary pivot matrix. Given $\pi \in \{0, \dots, n-1\}$ define the elementary pivot matrix

$$\tilde{P}(\pi) = \begin{pmatrix} e_{\pi}^T \\ e_1^T \\ \vdots \\ e_{\pi-1}^T \\ e_0^T \\ e_{\pi+1}^T \\ \vdots \\ e_{n-1}^T \end{pmatrix}$$

or, equivalently,

$$\tilde{P}(\pi) = \begin{cases} I_n & \text{if } \pi = 0 \\ \left(\begin{array}{c|c|c|c} 0 & 0 & 1 & 0 \\ 0 & I_{\pi-1} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{n-\pi-1} \end{array} \right) & \text{otherwise,} \end{cases}$$

where n is the size of the permutation matrix.

When $\tilde{P}(\pi)$ is applied to a vector, it swaps the top element with the element indexed with π . When it is applied to a matrix, it swaps the top row with the row indexed with π . The size of matrix $\tilde{P}(\pi)$ is determined by the size of the vector or the row size of the matrix to which it is applied.

In discussing LU factorization with pivoting, we will use elementary pivot matrices in a very specific way, which necessitates the definition of how a sequence of such pivots are applied. Let p be a vector of integers satisfying the conditions

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{k-1} \end{pmatrix}, \quad \text{where } 1 \leq k \leq n \text{ and } 0 \leq \pi_i < n-i, \quad (5.3.1)$$

then $\tilde{P}(p)$ will denote the sequence of pivots

$$\tilde{P}(p) = \begin{pmatrix} I_{k-1} & 0 \\ 0 & \tilde{P}(\pi_{k-1}) \end{pmatrix} \begin{pmatrix} I_{k-2} & 0 \\ 0 & \tilde{P}(\pi_{k-2}) \end{pmatrix} \cdots \begin{pmatrix} 1 & 0 \\ 0 & \tilde{P}(\pi_1) \end{pmatrix} \tilde{P}(\pi_0).$$

(Here $\tilde{P}(\cdot)$ is always an elementary pivot matrix "of appropriate size.") What this exactly does is best illustrated through an example:

Example 5.3.2.3. Let

$$p = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 0.0 & 0.1 & 0.2 \\ 1.0 & 1.1 & 1.2 \\ 2.0 & 2.1 & 2.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix}.$$

Evaluate $\tilde{P}(p)A$.

▼ [Solution](#)

$$\begin{aligned}
& \tilde{P}(p)A \\
&= \text{< instantiate >} \\
& \tilde{P}\left(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}\right) \begin{pmatrix} 0.0 & 0.1 & 0.2 \\ 1.0 & 1.1 & 1.2 \\ 2.0 & 2.1 & 2.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix} \\
&= \text{< definition of } \tilde{P}(\cdot) \text{>} \\
& \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & \tilde{P}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) \end{array} \right) \tilde{P}(2) \begin{pmatrix} 0.0 & 0.1 & 0.2 \\ 1.0 & 1.1 & 1.2 \\ 2.0 & 2.1 & 2.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix} \\
&= \text{< swap first row with row indexed with 2 >} \\
& \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & \tilde{P}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) \end{array} \right) \begin{pmatrix} 2.0 & 2.1 & 2.2 \\ 1.0 & 1.1 & 1.2 \\ 0.0 & 0.1 & 0.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix} \\
&= \text{< partitioned matrix-matrix multiplication >} \\
& \left(\begin{array}{c|c} (2.0 & 2.1 & 2.2) \\ \hline \tilde{P}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) \begin{pmatrix} 1.0 & 1.1 & 1.2 \\ 0.0 & 0.1 & 0.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix} \end{array} \right) \\
&= \text{< swap current first row with row indexed with 1 relative to that row >} \\
& \left(\begin{array}{c|c} (2.0 & 2.1 & 2.2) \\ (0.0 & 0.1 & 0.2) \\ \hline \tilde{P}((1)) \begin{pmatrix} 1.0 & 1.1 & 1.2 \\ 3.0 & 3.1 & 3.2 \end{pmatrix} \end{array} \right) \\
&= \text{< swap current first row with row indexed with 1 relative to that row >} \\
& \left(\begin{array}{c|c} (2.0 & 2.1 & 2.2) \\ (0.0 & 0.1 & 0.2) \\ \hline (3.0 & 3.1 & 3.2) \\ (1.0 & 1.1 & 1.2) \end{array} \right) \\
&= \\
& \begin{pmatrix} 2.0 & 2.1 & 2.2 \\ 0.0 & 0.1 & 0.2 \\ 3.0 & 3.1 & 3.2 \\ 1.0 & 1.1 & 1.2 \end{pmatrix}
\end{aligned}$$

The relation between $\tilde{P}(\cdot)$ and $P(\cdot)$ is tricky to specify:

$$\tilde{P}\left(\begin{pmatrix} \pi_0 \\ \pi_1 \\ \vdots \\ \pi_{k-1} \end{pmatrix}\right) = P\left(\begin{pmatrix} I_{k-1} & 0 \\ 0 & \tilde{P}(\pi_{k-1}) \end{pmatrix}\right) \cdots \begin{pmatrix} 1 & 0 \\ 0 & \tilde{P}(\pi_1) \end{pmatrix} \tilde{P}(\pi_0) \begin{pmatrix} 0 \\ 1 \\ \vdots \\ k-1 \end{pmatrix}.$$